



GE Medical Systems

Technical Publications

Direction 2100617

Revision 1

Signa[®] Advantage[™] 1.0T Anterior Neck Surface Coil

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Operating Documentation

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3/12/92



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<u>REVISION</u>	<u>DATE</u>	<u>PRIMARY REASON FOR CHANGE</u>
REV 0	11/15/93	Initial Release
REV 1	6/18/96	Add 60cm values to Table 2-2.

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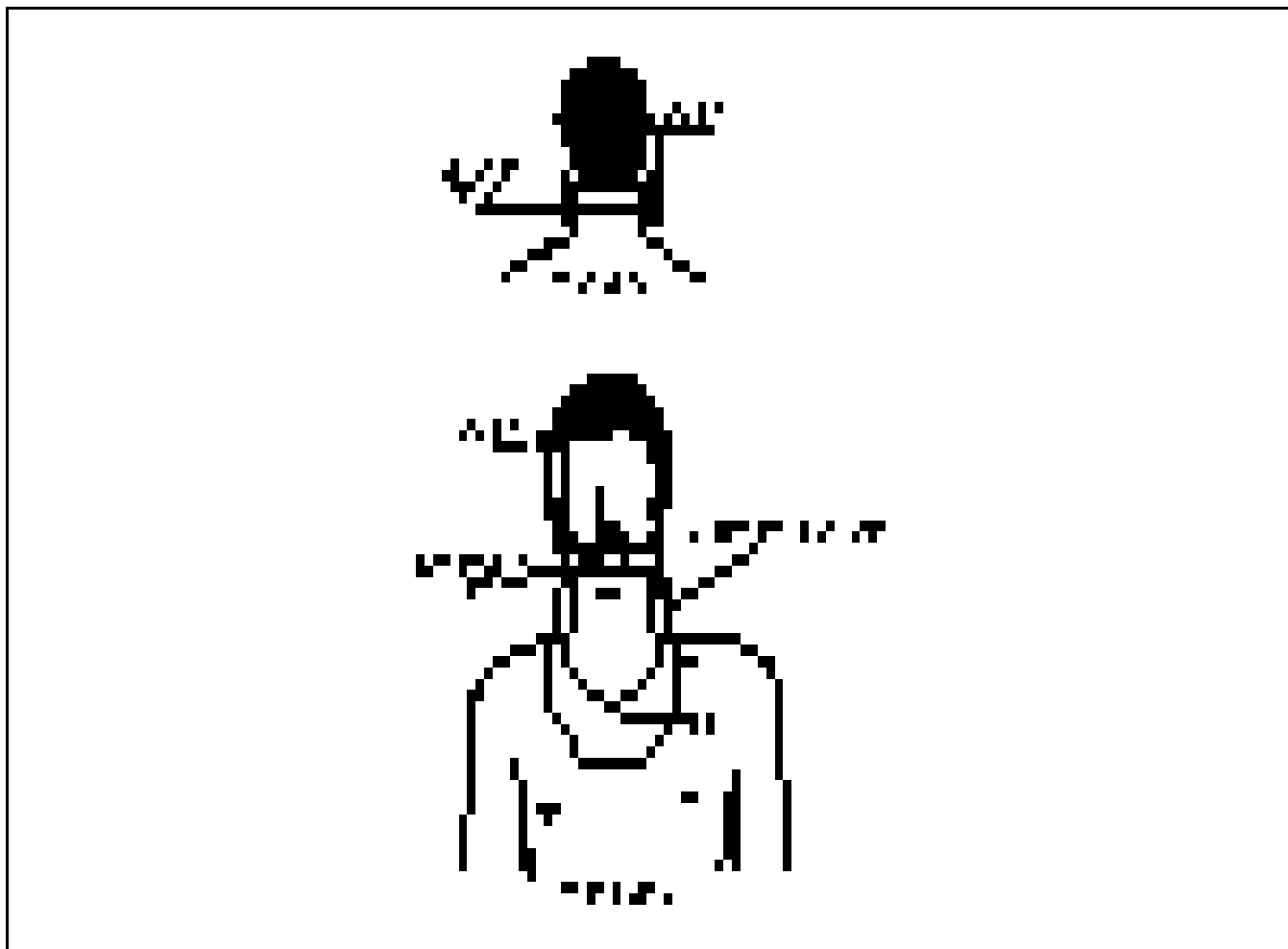
SECTION 1 - INTRODUCTION

1-1 THE ANTERIOR NECK SURFACE COIL

The Anterior Neck Coil is to be used with the Signa 1.0T system. Coil connection to the scanner Head Coil Port will be with an integral coil cable. See ILLUSTRATION 1-1.

Patient scan times can vary from a few minutes to an hour. To reduce factors, such as perspiration, and restriction of peripheral vascular circulation, the outer housing size is minimized. An opening in the housing is provided to promote air circulation, thus allowing venting and operator landmarking. Disposable liners are available. These liners, a flat sheet of nonabsorbent blue drape material with an adhesive strip attached, cover the area that comes in close contact with the nose and mouth.

Straps are provided to secure the coil to the patient. These straps employ Velcro® material allowing easy adjustment of the coil for different patient sizes, patient comfort, and quick removal.



ANTERIOR NECK COIL
ILLUSTRATION 1-1

1-2 COMPATIBILITY

The Anterior Neck Surface Coil is compatible with the following hardware configurations:

- GE SIGNA ADVANTAGE™ 1.0T Systems

1-3 RELATED DOCUMENTS

- Direction 15400 - Signa Advantage 1.5T, 1.0T, and 0.5T Systems
- Direction 15404 - Signa Advantage 1.5T, 1.0T, and 0.5T Renewal Parts.

1-4 ORGANIZATION OF THIS DOCUMENT

This document is divided into the following sections:

Section 1, INTRODUCTION, describes what the Anterior Neck Surface Coil is and where the Anterior Neck Surface Coil can be used.

Section 2, SET UP AND CALIBRATION, describes installation procedures.

Section 3, FUNCTIONAL CHECKS, describes normal power-up sequence.

Section 4, REPLACEMENT/MAINTENANCE, describes field maintenance procedures

Section 5, RENEWAL PARTS, lists field replaceable service kits and parts.

SNR Data Table

Coil Return Form

TLT Table

1-5 ENVIRONMENTAL REQUIREMENTS

Operate and store the Anterior Neck Surface Coil in the Scanner Room.

SECTION 2 - SET UP AND CALIBRATION

2-1 CHECKING THE SHIPPING LIST

Table 2-1 lists the M1885AK Signa Anterior Neck Coil parts. Check that all parts have been shipped.

TABLE 2-1
COIL PARTS

QTY.	ITEM	PART NO.
1	Anterior Neck — Medium Coil	46-328291P1
1	Disposable Liners	N/A
1	Service Manual	2100617

2-2 INSTALLING THE ANTERIOR NECK COIL

At the boot Terminal, install a new soft key. This key will be used by the operator to select Anterior Neck imaging. Name the key “**Ant Neck**”.

Refer to the Signa Advantage 1.5T, 1.0T, & 0.5T Systems Manuals for information on installing soft keys (use Coil default values in Table 2-2)

TABLE 2-2
COIL VALUES

SYSTEM	COIL NAME	COIL TYPE	EXTREMITY COIL	CABLE LOSS	COIL LOSS	RECON SCALE FACTOR	XMIT COIL TYPE	XMIT ATTN	MULTI-COIL.
1.0T	ANT NECK	surface	no	1.24	0.103 - 55cm 0.549 - 60cm	Head Coil Recon Scale Factor x 1.00	Quad	0	no

2-3 FUNCTIONAL CHECKS

1. Perform a Body coil scanner performance verification. Refer to Section 3-1, SIGNA ADVANTAGE 1.0T SCANNER VERIFICATION.
2. Perform an Anterior Neck Coil performance verification. Refer to Section 3-2, ANTERIOR NECK COIL IMAGING PERFORMANCE VERIFICATION.
3. Perform an SNR Image analysis. Refer to Section 3-3, SNR IMAGE ANALYSIS.

2-4 PERIODIC QUALITY ASSURANCE CHECK

On a periodic basis, such as during planned maintenance, perform the quality assurance check outlined below to ensure that the Anterior Neck Coil is operating properly with no appreciable degradation of image quality.

1. Check the external cable for cracks or breaks once each week.
2. Perform an Anterior Neck Coil performance verification. Refer to Section 3-2, ANTERIOR NECK COIL IMAGING PERFORMANCE VERIFICATION.
3. Record the date and value calculated in SNR QA DATA TABLE
4. As instructed in the Data Table, divide the SNR value obtained in the periodic QA check by the original SNR value.
5. If this ratio is not greater than 85%, then there may be a problem with the Anterior Neck Coil or the MR system. Contact the GE Service Representative.

SECTION 3 - FUNCTIONAL CHECKS

3-1 SIGNA ADVANTAGE™ 1.0T SCANNER VERIFICATION

Note

An alternative proprietary procedure is available for GE use and to customers with a valid Advanced Service Package Limited License. Refer to "TLT PROCEDURE" located in Direction 15491, *Signa Advantage 1.5T, 1.0T, and 0.5T system Troubleshooting Guide*.

1. Scanner performance in the body coil mode should be verified. The RF power level calibration from the body coil mode is used to verify the surface coil decoupling. Landmark the 100 mm TLT phantom without the coil. Place the TLT Phantom on the Universal Phantom Holder.
2. Set up the scan using the parameters in TABLE 3-1. Coil Type: **[Body Coil]** and Number of Echos: **[4]** should be inserted. Perform an **Auto Prescan** to calibrate the RF power level. Record the value for TG, R1, and R2.
3. Run the scan. Observe the resulting images. Ensure that there are no artifacts of any sort on any of the images.
4. Select **[Cancel]**, **[New Series]**, to perform SNR comparisons. Acquire two identical images using the previously loaded scan parameters in TABLE 3-1 after inserting Number of Echos: **[1]**. Immediately after acquiring the

TABLE 3-1
SCAN PARAMETERS

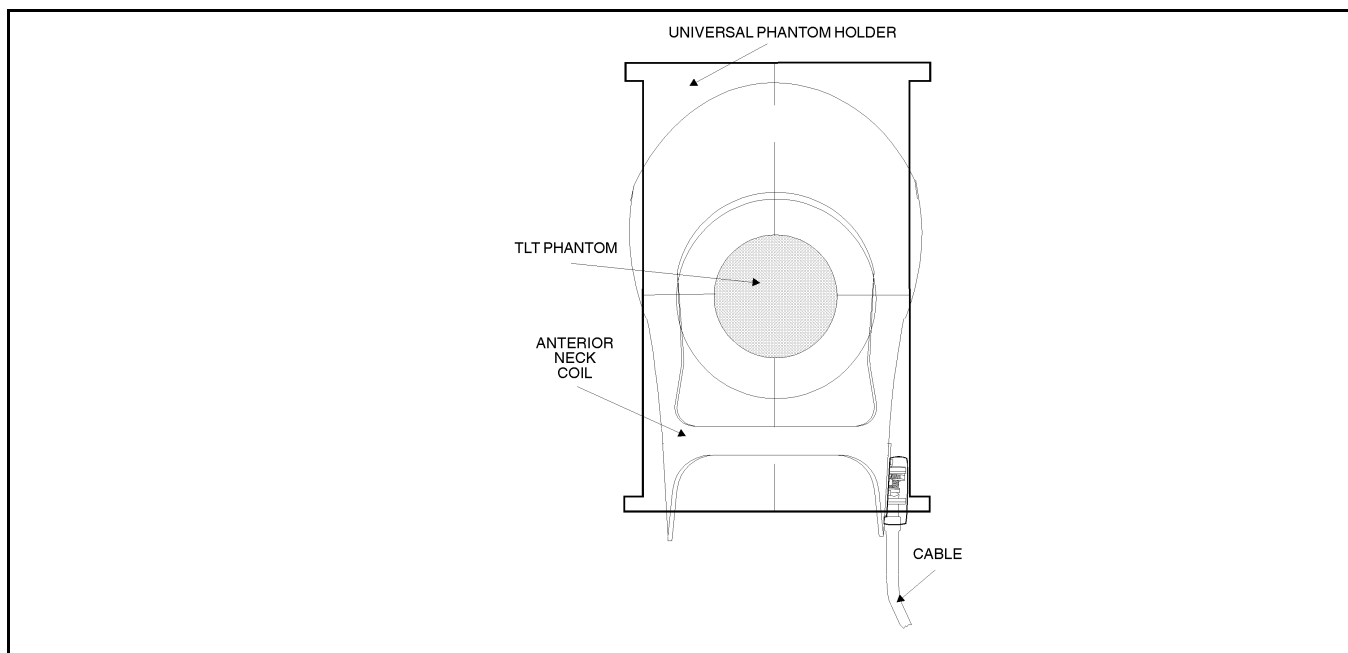
ID:	geservice	Auto CF:	[Peak]
Name:		Receive Bandwidth:	[16kHz]
Weight (lb):	111	Field of View:	[24 cm]
Patient Entry:	[Head First]	Slice Thickness:	[10 mm]
Patient Position:	[Supine]	Interscan Spacing:	[0]
Axial/Sag Landmark:	[Sternal Notch]	Start Location (I/S):	0
Coil Type:	[++] See Note	End Location (I/S):	0
Scan Plane:	[Axial]	No of Scan Locations ¹	
Image mode:	[2D]	FOV Center (L/R):	0 (P/A): P90
	[Monitor SAR]	Acq. Matrix: (freq.)	[256]
Pulse Sequence:	[Spin Echo]	Acq. Matrix: (phase)	[128]
Imaging Options:	[None]	Frequency Direction:	[R/L]
Number of Echos:	[*] See Note	Phase FOV:	default
Echo Time (TE):	[20 ms]	Imaging Time:	[1 Nex]
Rep Time (TR):	[Other] 600 ms	Contrast:	[No]
		Table Delta	0

NOTE: * **[4]** for Section 3-1-2; 3-2-2
[1] for Section 3-1-4; 3-2-4
++ **[Body]** for Section 3-1-2; 3-2-2
[Other][Ant Neck] for Section 3-1-4; 3-2-4

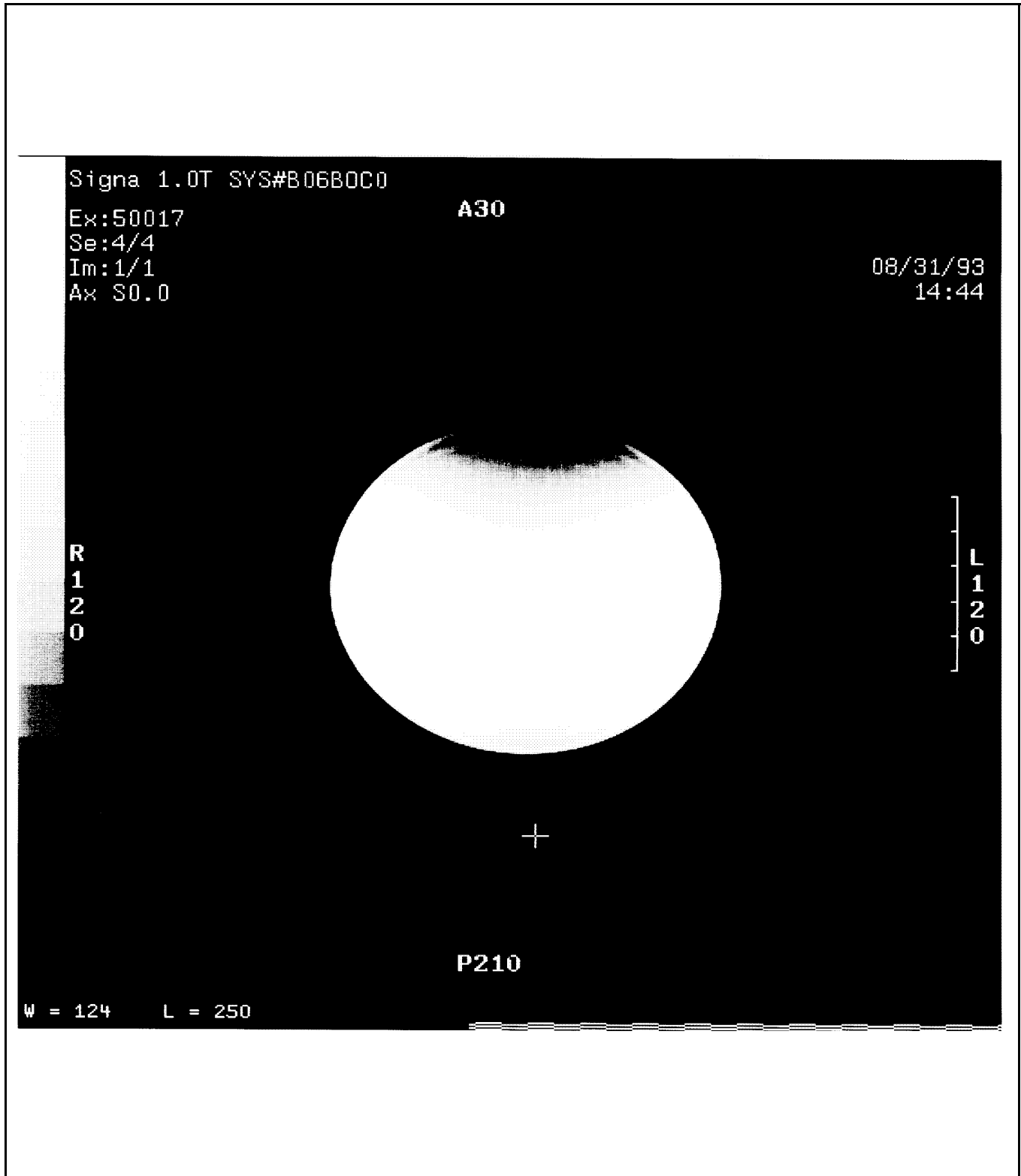
first image, select **Scan** to obtain a second. Note the series number for future reference.

3-2 ANTERIOR NECK COIL IMAGING PERFORMANCE VERIFICATION

1. Set up the Anterior Neck coil and 100 mm TLT phantom as shown in ILLUSTRATION 3-1. Connect the surface coil cable to the **SURFACE COIL QUICK DISCONNECT BOX** and connect to the **HEAD COIL PORT**.
2. Select [**Cancel**], [**New Series**], and repeat the scan using the parameters in TABLE 3-1. Coil Type: [**Other**][**Ant Neck**] and Number of Echos: [**4**] should be used. Use **Auto Prescan** to set the values of R1 and R2 to those in step 3-1-2. If the value of TG does not match the value obtained in step 3-1-2, use **Manual Prescan** to set the value of TG.
3. Run the scan. Ensure that there are no artifacts of any sort on any of the images. Study the image using a **Window Width** and **Window Level** to obtain good contrast. A properly functioning coil will produce an image with a smooth signal pattern, gradually dropping in intensity as the distance from the coil increases, see ILLUSTRATION 3-2. A decoupling failure will cause holes, distortions, or striped patterns in the image, Refer to Section 3-4 to 3-7 to troubleshoot decoupling failures.
4. Repeat step 2 with Number of Echos: [**1**]. Run the scan. Immediately after acquiring the first image, select **Scan** to obtain a second.
5. Compute the SNR using the procedure described in Section 3-3, SNR IMAGE ANALYSIS. Compare the signal to noise ratio of the first echo of the Anterior Neck Coil images with the signal to noise ratio of the first echo of the body coil images of step 3-1-4. The Anterior Neck Coil scans should display a signal to noise ratio greater than 2.5 times that of the body coil scans.
6. Inspect the images for visible ghosting (similar in appearance to motion artifacts) in the phase encoding direction. If images contain ghosting artifacts, suspect an intermittent cable or connection. Refer to Section 3-4 to 3-7 for troubleshooting information.



TLT PHANTOM PLACEMENT
ILLUSTRATION 3-1



NORMAL DECOUPLING
ILLUSTRATION 3-2

3-3 SNR IMAGE ANALYSIS

Description

The SNR tool retrieves two operator selected images. Signal value is computed as the mean pixel value in a ROI covering 80% of the image. The image is analyzed to determine the center for positioning the ROI. A difference image is created by subtracting the second image from the first and to calculate noise from the subtracted image. Signal value, noise value, and SNR are reported. The difference image can be saved with the results annotated (option).

Procedure

1. Touch **[Utilities]**, **[MR Tools]**, **[Image Quality]**, then **[SNR Test]**. The SNR Test screen is displayed. See ILLUSTRATION 3-3.
2. Enter first image exam, series, and image numbers. If exam, series, or image numbers are not known, select **[List Exams]**, **[List Series]**, or **[List Images]** to display list to choose from. Enter second image series and image number. The second image Exam Number is defaulted to the same as first image.

Note

Image number selection must be back lit (highlighted) to be able to enter information. Use Switch key on keyboard to transfer control from left to right side of Touch Screen.

3. Select **[Filter Off]** and **[Image Off]**. Touch **[Accept]** to begin analysis. The final values are displayed on the screen.
4. Record Signal to Noise Ratio on SNR QA DATA TABLE. Touch **[Continue]**, select the next exam and repeat for each pair.

NEMA STANDARD SNR TEST				
	Filter Off	Image Off		
First Image	List/Sel Exams	List/Sel Series	List/Sel Images	
Second Image		List/Sel Series	List/Sel Images	
Cancel Utility Main MR Tools Main Image Quality Accept				

Note: Accept changes to continue only after an analysis has been performed.

SNR TEST SCREEN
ILLUSTRATION 3-3

3-4 COIL IMPEDANCE MEASUREMENTS

1. Move to an area where the magnetic field from the Signa Advantage 1.0T magnet is less than 5 Gauss. Load the surface coil with a human subject. Use the procedure in the Operation Manual to place the surface coil.
2. Connect the surface coil cable to the vector impedance meter probe using an appropriate adapter. Use the vector impedance meter to measure the coil impedance at 42.68 MHz. Ensure that the cable does not touch or pass close to the coil housing while making this measurement.
3. The impedance magnitude should nominally be 50 ohms under loaded conditions. It should fall within the range of 40 to 60 ohms under test conditions.
4. The phase angle should nominally be zero degrees. It should fall within the range of -20 to +20 degrees under test conditions.

3-5 COIL INTEGRITY TESTS

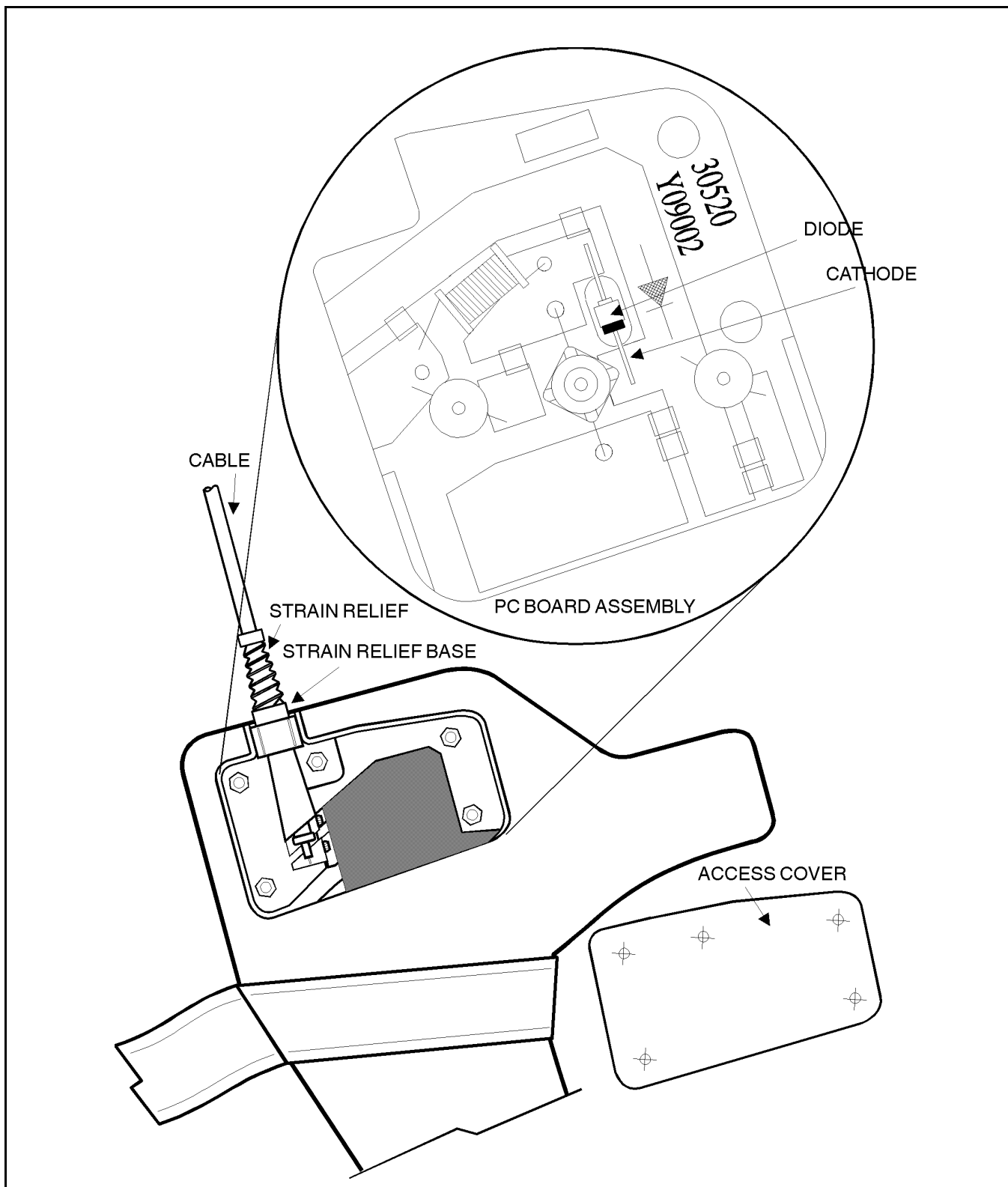
1. Repeat Section 3-4-1 while flexing the surface coil cable, especially at the fitting and the coil housing strain relief. Using the **HIGH SPEED** mode on the vector impedance meter, note any erratic fluctuations in the coil impedance. Any erratic or unstable readings may indicate a defective cable or connector. Refer to Section 4, REPLACEMENT/MAINTENANCE.

3-6 DECOUPLING DIODE TESTS

1. Use the digital multimeter on the diode test range to measure across the coil connector. With the negative lead to the connector shell and the positive to the connector center pin, the reading should be between 0.4 and 0.9. Reversing the leads should cause the reading to be infinity (overrange).
2. An open reading in both directions indicates an open diode, or an open cable. Refer to Section 3-7 to check the cable. Refer to Section 4-3, DIODE REPLACEMENT to replace the diode.
3. A short circuit or excessively low reading indicates a shorted diode or a shorted cable. Refer to Section 3-7 to check the cable. Refer to Section 4-3, DIODE REPLACEMENT to replace the diode.

3-7 CABLE TESTS

1. If the surface coil passes the tests in Sections 3-4, 3-5, and 3-6, the cable can be assumed to be functional.
2. If a short or open circuit was discovered in Section 3-6, the cable may be tested by unsoldering the cable at the PC board via the service access cover, as shown in ILLUSTRATION 3-4. The cable may then be tested for continuity of the center conductor and shield, and for shorts between the center conductor and shield using the multimeter continuity test function.

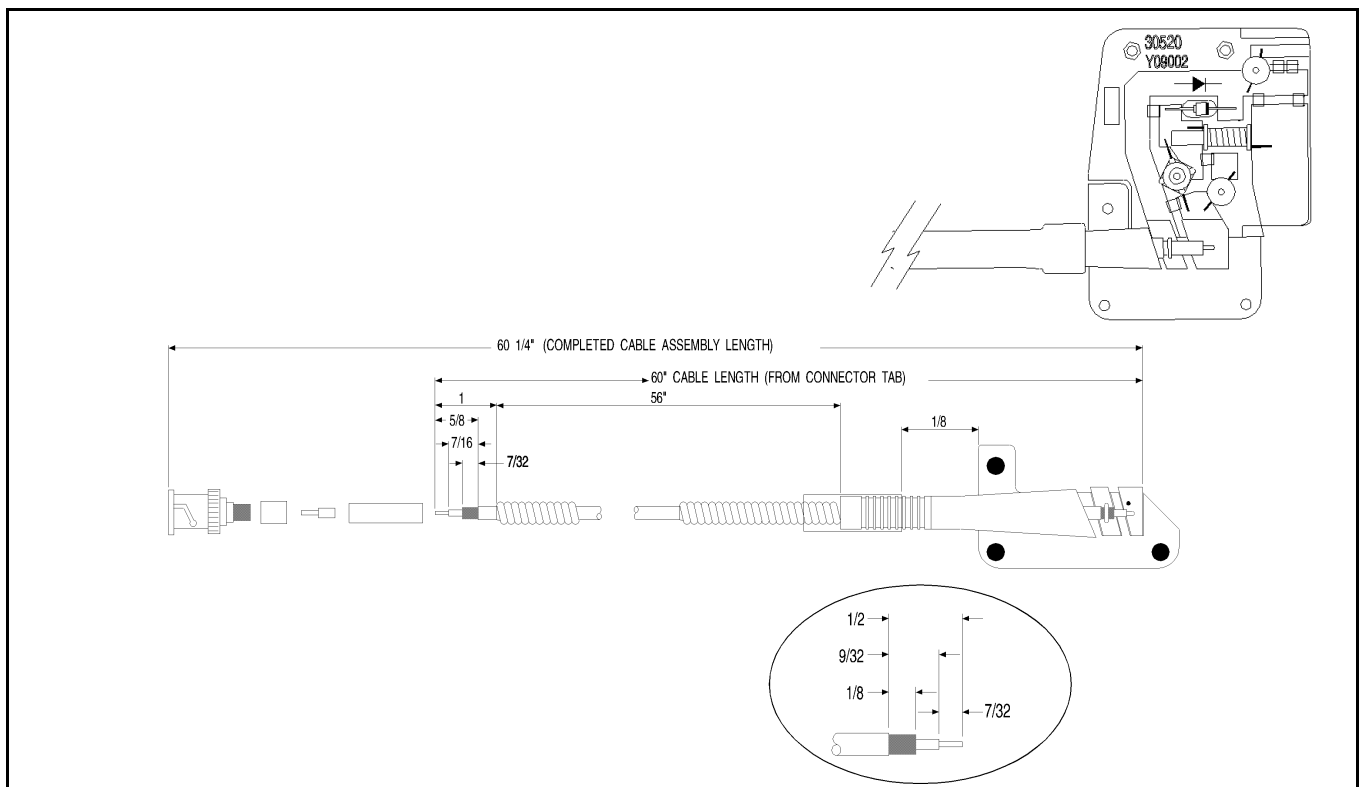


ACCESS COVER, INTERNAL CABLE CONNECTION
ILLUSTRATION 3-4

SECTION 4 - REPLACEMENT/MAINTENANCE

4-1 COAXIAL CABLE REPLACEMENT

1. Remove the service access cover from the housing by removing the nylon screws attaching the cover.
2. Unscrew the three stand-offs from the cable/PCB assembly.
3. Scrape off the conformal coating from two soldered connection tabs which join the cable/PCB assembly to the decoupling board.
4. Unsolder the connection tabs and remove cable/PCB assembly.
5. Place the new cable/PCB assembly on the screwposts. Do not tighten the screws.
6. Solder the tabs, then tighten the screws.
7. Attach the service access cover to the coil housing.
8. Repeat the tests in Sections 3-2 and 3-3 to verify proper coil operation.



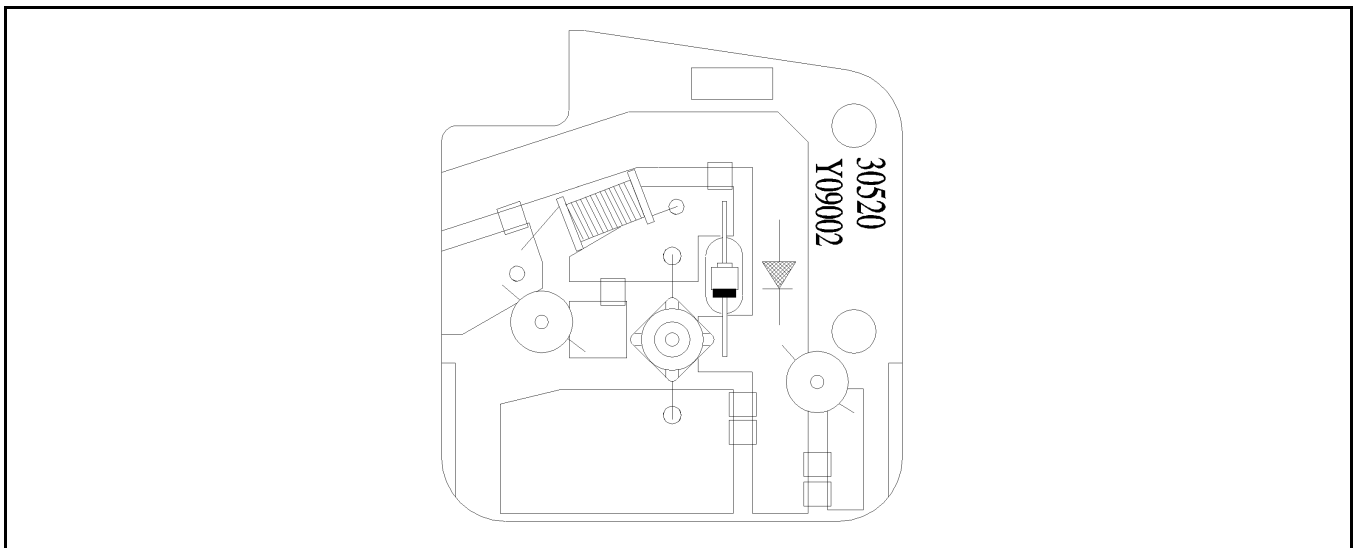
COAXIAL CABLE REPLACEMENT
ILLUSTRATION 4-1

4-2 OUTPUT CONNECTOR REPLACEMENT

1. If the cable will be shortened to less than 57-1/8 inches by a repair, it is recommended that the entire cable be replaced in the event of a connector or connector-end cable failure.
2. Replace the connector using ILLUSTRATION 4-1 as a guide. Use a new BNC coaxial fitting - part number 46-271494P1, and crimp tool - part number 46-255841, with insert - part number 46-255841P100, to install the replacement connector on the cable.
3. Repeat the tests in Sections 3-2 and 3-3 to verify proper coil operation.

4-3 DIODE REPLACEMENT

1. If it has been determined that a diode must be replaced, remove the access cover. See ILLUSTRATION 4-2.
2. Unsolder defective diode and replace with new diode - part number 46-221735P1.



DIODE PLACEMENT
ILLUSTRATION 4-2

4-4 REPLACING THE MECHANICAL HARDWARE

1. Replace the entire Anterior Neck Coil. Refer to Section 5, RENEWAL PARTS, for Anterior Neck Coil part number.

4-5 CHECKING THE EXTERNAL CABLE

1. Check the external cable for cracks or breaks once each week. Replace the external cable if any damage or wear is found. Refer to Section 4-2, OUTPUT CONNECTOR REPLACEMENT, for replacement procedure.

4-6 CLEANING THE ANTERIOR NECK COIL

1. Clean the Anterior Neck Coil and external cable with a mild dishwashing liquid and water solution. Wet a soft cloth with the solution and proceed to clean.

SECTION 5 - RENEWAL PARTS

5-1 SERVICE KITS

1. A service kit is available for replacement of the cable assembly:

CABLE KIT 2100937 — Consisting of the following items:

ITEM	DESCRIPTION	QTY
1	Cable Assembly	1
2	Screws	5

2. The Anterior Neck Coil Part number is:

ITEM	DESCRIPTION	QTY
1	Anterior Neck Coil with Cable—46-328291P1	1

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SNR QA DATA TABLE

Use the space provided below to record the calculated signal to noise ratio—SNR, data obtained from Section 4, SNR IMAGE ANALYSIS. After recording the SNR data obtained during the initial coil installation, record subsequent SNR data in column 2, enter the date in column 1 as a periodic QA check. If the ratio of any of the coils found is not greater than 85%, then there is a problem in the coil or the MR system.

Original SNR data obtained at initial coil installation:

Body Coil SNR Value: _____

Anterior Neck SNR value: _____

Date: _____

SNR DATA QA (QUALITY ASSURANCE) CHECK:

1	2	3	
Date	Ant. Neck Coil QA SNR Value:	Is new value divided by original value >85%?	
_____	_____	_____	Y/N
_____	_____	_____	Y/N
_____	_____	_____	Y/N
_____	_____	_____	Y/N
_____	_____	_____	Y/N
_____	_____	_____	Y/N
_____	_____	_____	Y/N
_____	_____	_____	Y/N
_____	_____	_____	Y/N
_____	_____	_____	Y/N
_____	_____	_____	Y/N

Make additional copies of this document as needed.

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DIRECTION 2100617

DEFECTIVE SURFACE COIL RETURN FORM

NOTE

For proper assessment of defective returned coils, both sides of this form should be completely filled out and accompany all returned coils. Include films or prints of any image quality complaints with a description of the scan prescription used.

DATE: _____

SITE NAME: _____

SITE ADDRESS: _____

SERVICE ENGINEER: _____

COIL SERIAL NUMBER: _____

DATE COIL INSTALLED: _____

DESCRIPTION OF COIL PROBLEM: _____

(Please be descriptive; "Broken" is not enough) _____

ELECTRICAL CHECKS	
VECTOR IMPEDANCE METER TEST - COIL LOADED WITH HUMAN SUBJECT IN FREE SPACE—Refer to Section 3-4-1.	
MAGNITUDE	
PHASE	
VECTOR IMPEDANCE METER TEST - COIL UNLOADED IN FREE SPACE—Refer to Section 3-4-1.	
MAGNITUDE	
PHASE	
PIN DIODE TEST—Refer to Section 3-4-3.	
DIODE DROP—FORWARD BIAS	
DIODE DROP—REVERSE BIAS	

Note

An alternate proprietary procedure is available to GE use and to customers with a valid Advanced Service Package Limited License. Refer to "TLT PROCEDURE" located in Direction 15491, *Signa Advantage 1.5T, 1.0T & 0.5T System Troubleshooting Guide*. Use the table to record the TLT results of the defective coil.

TABLE DR-1
TLT DATA FOR DEFECTIVE SURFACE COIL

SITE:		NAME:	DATE:
TLT FILE NUMBER:			
SNR:	NOISE		
	SNR MEAN		
	SNR SIGNAL		
	SNR AREA		
TRMAP:	89-91	%	%
	85-95	%	%
	65-115	%	%
	FLIP MEAN		
	FLIP SDV		
	RCV MEAN		
	RCV SDV		
	DIFF MEAN		
	DIFF SDV		



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